DSA210 Term Project: Relationship Between TikTok Popularity and Spotify Streams

Student: Zeynep Sezer
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1. Introduction

1.1 Motivation

In recent years, TikTok has become one of the most influential platforms in shaping global music trends. Many songs go viral on TikTok before becoming popular on music streaming platforms like Spotify. Because of this, it is interesting to study whether TikTok popularity actually has an effect on Spotify streaming numbers.

The main motivation of this project is to understand whether songs that become popular on TikTok also tend to get more streams on Spotify. This helps to explore how social media platforms influence music consumption behavior.

1.2 Objectives

The main objectives of this project are:

- To analyze the relationship between TikTok popularity and Spotify streaming numbers
- To clean and merge datasets from both platforms
- To explore correlations between TikTok engagement and Spotify streams
- To build machine learning models to predict Spotify streams using TikTok data
- To evaluate whether TikTok popularity can be used as an indicator of music success

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2. Data & Methodology

2.1 Data Sources

The dataset used in this project was created by combining two main sources: TikTok popularity data and Spotify streaming data.

The TikTok data includes information about how popular each song is on the platform, based on usage or engagement. The Spotify data contains the number of streams for each song, showing how often the songs are listened to on Spotify.

Since there is no single dataset that directly matches TikTok popularity with Spotify streams, the two datasets were merged manually using track names and artist information. Before merging, some basic cleaning was done to make sure the names matched properly, since there were differences in spelling and formatting.

After merging, a final dataset was created that includes both TikTok and Spotify information for each track. This combined dataset was then used to analyze the relationship between TikTok popularity and Spotify streaming numbers.

2.2 Data Processing

- Missing values were checked and handled
- Track names were standardized for better matching
- Spotify streams were log-transformed to reduce skewness
- Basic feature preparation was done for modeling

2.3 Data Analysis Methods

The following techniques were used in this project:

- Exploratory Data Analysis (EDA)
- Correlation analysis
- Linear Regression
- Random Forest Regression

These methods were used to both understand the data and test whether TikTok popularity can help predict Spotify streaming numbers.

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3. Exploratory Data Analysis (EDA)

In the first step of the analysis, the distribution of TikTok popularity and Spotify streams was examined.

Key observations:

- Spotify streams are highly skewed, meaning a small number of songs have very high stream counts
- TikTok popularity shows a similar pattern, where a few songs go viral
- There is a visible positive relationship between TikTok popularity and Spotify streams

These patterns suggest that viral songs tend to perform well across both platforms.

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4. Correlation Analysis

A correlation analysis was performed to measure the relationship between TikTok popularity and Spotify streams.

The results show a **positive correlation** between the two variables.

This means that songs that are more popular on TikTok generally tend to have higher Spotify streaming numbers. However, this relationship is not perfect, and other factors may also influence streaming success.

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5. Machine Learning Analysis

To further analyze the relationship, machine learning models were built to predict Spotify streams using TikTok popularity data.

5.1 Linear Regression

A linear regression model was used as a baseline model.

- The model showed a clear positive relationship between TikTok popularity and Spotify streams
- This suggests that TikTok popularity has predictive value

5.2 Random Forest Regression

A Random Forest model was also used to capture more complex relationships.

- The model performed better than linear regression
- TikTok popularity was one of the most important features in prediction

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5.3 Model Interpretation

Both models show that TikTok popularity is an important factor in predicting Spotify streaming numbers. However, the relationship is not perfect, meaning that other external factors also play a role.

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6. Discussion

The results of this project suggest that TikTok has a noticeable influence on music streaming behavior. Songs that become popular on TikTok are more likely to receive higher streaming numbers on Spotify.

However, this relationship should not be seen as fully causal. Some songs may already be popular before going viral on TikTok, and other factors such as marketing or playlists can also affect streaming numbers.

Overall, TikTok appears to act as a strong boosting platform for music discovery.

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7. Conclusion

This project analyzed the relationship between TikTok popularity and Spotify streaming numbers using statistical analysis and machine learning methods.

The results show a consistent positive relationship between the two platforms. This suggests that TikTok can be considered a useful indicator of Spotify streaming success, highlighting the strong connection between social media trends and music consumption behavior.

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8. Limitations & Future Work

Limitations:

- Some errors may exist in matching tracks between datasets
- The analysis focuses only on correlation, not full causality
- Other important factors (like marketing or genre) were not included

Future Work:

- Improve dataset matching using better identifiers
- Include additional features such as genre or release date
- Analyze trends over time using time-series data
- Expand dataset size for more robust results

AI Usage Disclaimer

In this project, I used AI tools to help with improving code structure, debugging, and organizing the report. AI was mainly used for:

- **Coding Support:** Assisting with debugging Python code, improving Pandas operations, and supporting model implementation steps.
- **Writing Support:** Helping to improve clarity in explanations of the analysis and results.

All data analysis, model building, interpretation of results, and final conclusions were reviewed and validated manually.

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10. References & Tools

Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn

Data Sources: TikTok popularity dataset, Spotify streaming dataset